

Course Name: Computer Networks and Internet Protocol

Assignment - Week 7 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1

What is the transmission delay for pushing a packet of size 1MB through a network with a bandwidth 16Mbps?

- a) 2 seconds
- b) 0.5 seconds
- c) 1 second
- d) 10 second

Correct Answer: (b)

Detailed Solution: Since network bandwidth is 16Mbps so for sending 8Mb of data through the network it will take 0.5 seconds.

QUESTION 2

File transfer applications require non-real-time variable bit rate (NRT-VBR).

Is this statement true or false?

- a) True
- b) False

Correct Answer: b False

Detailed Solution: File transfer is an available bit-rate (best-effort) application.

QUESTION 3

The primary goal of QoS is

- a) Degrade loss characteristics.
- b) Control jitter



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- c) Increase communication latency
- d) Reduce network security mechanisms

Correct Answer: (b)

Detailed Solution: The primary goal of QoS is to control jitter, improve loss characteristics and decrease communication latency.

QUESTION 4

Which of the following statements is TRUE?

- a) TCP is Inelastic Traffic and therefore not preferred for real-time video streaming.
- b) TCP is Elastic Traffic and therefore preferred for real-time video streaming.
- c) UDP is Elastic Traffic and therefore preferred for real-time video streaming.
- d) UDP is Inelastic Traffic and therefore preferred for real-time video streaming.

Correct Answer: (d)

QUESTION 5

Based on QoS requirements, video conferencing comes under which class of application?

- a) Constant bit rate
- b) Real-time variable bit rate
- c) Non-real-time variable bitrate
- d) Available bit rate or best-effort

Correct Answer: (b)

QUESTION 6

Which of the following is not a benefit of traffic shaping?

- a) Reducing network congestion
- b) Controlling the rate of traffic flow
- c) Improving network efficiency
- d) Prioritizing certain types of traffic

Correct Answer : (d)

Detailed Solution: Buffering packets to smooth out bursts of traffic is an example of a traffic shaping technique. It helps to ensure that the network does not become congested with sudden spikes in traffic.

QUESTION 7

Which of the following is an example of a traffic scheduling algorithm?

- a) Weighted Fair Queuing (WFQ)
- b) Leaky Bucket
- c) Token Bucket
- d) Random Early Detection (RED)

Correct Answer: (a)

Detailed Solution: Weighted Fair Queuing (WFQ) is an example of a traffic scheduling algorithm. It assigns weights to different types of traffic to ensure that they are delivered fairly based on their importance.

QUESTION 8

Consider the token bucket algorithm, with token arrival rate = packet arrival rate = r per second. Bucket size is b . What is the maximum burst size (MBS) and the output rate?

- a) rb and $r+b$
- b) No burst and rb
- c) No burst and r
- d) $r + b$ and rb

Correct Answer: (c)

Detailed Solution: Since the rate of token arrival is the same as the rate of packet arrival, the bucket will always have enough token to send the packets at the same rate they arrive. Therefore there will not be any burst.

QUESTION 9

Which of the following statements about Resource Reservation Protocol (RSVP) is FALSE?

- a) RSVP is difficult to implement on the internet because it requires coordination among intermediate routers.
- b) All the RSVP daemons in all the intermediate routers in an end-to-end path need to coordinate with each other.
- c) RSVP uses differentiated services fields in the IP header for traffic classification.
- d) RSVP is a protocol of the Integrated Service (IntServ) QoS architecture.



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Correct Answer: (c)

QUESTION 10

In a network with a maximum queue size of 100 packets and a minimum threshold of 40 packets, what is the drop probability when the current queue size is 70 packets and the maximum threshold is set to 90 packets using the RED algorithm?

- a) 20%
- b) 40%
- c) 60%
- d) 80%

Correct Answer: (a)

Detailed Solution: The RED algorithm calculates the drop probability based on the current queue size and the thresholds set for the queue. In this case, the minimum threshold is 40 packets and the maximum threshold is 90 packets, so the range is 50 packets. When the current queue size is 70 packets, the midpoint of the range is 65 packets $[(40 + 90) / 2]$. The drop probability is calculated as a function of the distance from the midpoint, using a curve that starts at 0% when the queue is at the minimum threshold and reaches 100% when the queue is at the maximum threshold. At a distance of 5 packets from the midpoint, the drop probability is typically around 20%, so the answer is A.
